Hugo Marquerie 12/06/2026

Lem Desigualdad Triangular Producto Hiperbolico

Lema 1. *Sean* $x, y, z \in [0, 1)$*. Entonces,* $x \leqslant \dfrac{y+z}{1+yz} \implies \dfrac{1+x}{1-x} \leqslant \dfrac{1+y}{1-y} \cdot \dfrac{1+z}{1-z}$.

***Demostración*:** Consideramos $\varphi\colon \mathbb{R} \setminus \{1\} \to \mathbb{R}$ dada por $\varphi(t) = \frac{1+t}{1-t}$. Entonces, $\forall t < 1 :$ $\varphi'(t) = \frac{2}{(1-t)^2} > 0$, por lo que φ es estrictamente creciente en $(-\infty, 1)$.

Por tanto, como $x, y, z \in [0, 1)$ y $x \leqslant \frac{y+z}{1+yz}$, tenemos que

$$\varphi(x) \leqslant \varphi\left(\frac{y+z}{1+yz}\right) \implies \frac{1+x}{1-x} \leqslant \frac{1+\frac{y+z}{1+yz}}{1-\frac{y+z}{1+yz}}.$$

Operando en el lado derecho, llegamos a que

$$\frac{1+\frac{y+z}{1+yz}}{1-\frac{y+z}{1+yz}} = \frac{(1+yz)+(y+z)}{(1+yz)-(y+z)} = \frac{(1+y)(1+z)}{(1-y)(1-z)} = \frac{1+y}{1-y} \cdot \frac{1+z}{1-z} = \varphi(y) \cdot \varphi(z).$$

Esto implica la desigualdad que queríamos demostrar. ■

Referenciado en

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